Supplementary material: The 1.0–1.2 TeV Thermal Higgsino Against Planck, LZ, XENONnT and Gamma-Ray Limits: A Toy-Model Study

HEP-Genesis-Agent session record
Project: ramp-combo-higgsino-dd

2026-10-16

Supplementary material, generated mechanically from the session record: it lists exactly what ran, in order, without interpretation.

A Implementation details

This appendix is generated mechanically from the session record; it lists exactly what ran, in order, without interpretation.

A.1 Session

- Source: cli client; conversation `b128dd5f`
- Started: 2026-10-16 14:31:17; report generated: 2026-10-16 15:02:03
- Project: ramp-combo-higgsino-dd (`/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-higgsino-dd`)
- Compute execution site(s) selected in the session: local
- 1 user turn(s), 89 tool call(s) (0 returned errors), 0 figure(s), 0 data file(s)

A.2 Models

Stage	Model	Endpoint
quick	`claude-opus-5-5`	`https://mag.genesis.americansciencecloud.org/v1`
report	`claude-opus-5-5`	`https://mag.genesis.americansciencecloud.org/v1`
coder	`claude-opus-5-5`	`https://mag.genesis.americansciencecloud.org/v1`

A.3 MCP servers

Server	Transport	Location
hep-research-aid	stdio	`python -m hep_genesis.domain.mcp_server`
alcf-tools	stdio	`python -m hep_genesis.iri.alcf.mcp_server`
nersc-tools	stdio	`python -m hep_genesis.iri.nersc.mcp_server`

A.4 Tool-call log

Chronological; arguments and results are truncated for space (the full record is in the project's `provenance/` folder).

#	Tool	Arguments	Outcome
1	`serve-sbbn.sh list_sterile_nu_skills`	`{}`	`{ "status": "ok", "files": [], "message": "2 skills. Load one with load_sterile_nu_skill.", "metadata": { "skills": { "kev-sterile-dm-xray": "keV sterile-neutrino dark matter \u2014 ...`
2	`serve-hydro.sh list_observables`	`{}`	`{ "status": "success", "message": "14 observables available: GSMF, HMF, fGas, Pk-ratio, CSFR, CGD, CGED, CPP, CTP, CEP, CEEP, CMP, CYP, Pk_GO.", "files": [], "metadata": { "observables": {...`
3	`hep-research-aid list_agent_files`	`{}`	`[]`
4	`hep-research-aid list_input_files`	`{}`	`{ "status": "success", "files": [], "usage": "Pass any of these filenames to load_table_data()." }`
5	`hep-research-aid list_datasets`	`{}`	`{'count': 0, 'datasets': []}`
6	`hep-research-aid list_files`	`{}`	`Files in .: - .git - .gitignore - ISSUES.md - SCOPE.md - docs - downloads - logs - projects - reports - research-results - runs - scripts - venvs - workspace`
7	`hep-research-aid get_dispatch`	`{}`	`Dispatch: local execution.`

#	Tool	Arguments	Outcome
8	local search_arxiv_ abstracts	{"query": "higgsino dark matter thermal relic 1.1 TeV direct detection", "max_ results": 8}	Search failed: Page request resulted in HTTP 503 (https:// export.arxiv.org/api/ query?search_query= higgsino+dark+matter+thermal+relic+1 list=&sortBy= relevance&sortOrder= descen...
9	local search_arxiv_ abstracts	{"query": "pure higgsino inelastic scattering mass splitting xenon detectors", "max_results": 8}	Search failed: Page request resulted in HTTP 429 (https:// export.arxiv.org/api/ query?search_query= pure+higgsino+inelastic+scattering+ma list=&sortBy= relevance&sortOrder=...
10	local search_arxiv_ abstracts	{"query": "higgsino Sommerfeld enhancement relic abundance electroweak", "max_results": 6}	Search failed: Page request resulted in HTTP 429 (https:// export.arxiv.org/api/ query?search_query= higgsino+Sommerfeld+enhancement+reli list=&sortBy= relevance&sortOrder= descen...
11	hep-research-aid search_arxiv	{"query": "higgsino dark matter thermal relic direct detection", "max_results": 10}	Found 10 papers matching query: 'higgsino dark matter thermal relic direct detection' (source: OpenAlex) Paper 1: arXiv ID: 1611.00804 Title: The last refuge of mixed wino-Higgsino dark matter Author...
12	local web_search	{"query": "higgsino 1.1 TeV thermal relic inelastic dark matter LZ XENONnT arXiv"}	Sep 1, 2026 · The Higgsino dark matter of the minimal supersymmetric standard model is one of the most minimal and canonical WIMP candidates. Sep 2, 2026 · This Supplemental Material details the compu...

#	Tool	Arguments	Outcome
13	hep-research-aid search_arxiv	{"max_results": 10, "query": "inelastic higgsino LZ nuclear recoil excess mass splitting keV"}	Found 10 papers matching query: 'inelastic higgsino LZ nuclear recoil excess mass splitting keV' (source: OpenAlex) Paper 1: arXiv ID: 2609.09136 Title: A Warped Extra Dimensional Candidate for the L...
14	hep-research-aid search_arxiv	{"max_results": 8, "query": "wino higgsino thermal mass Sommerfeld bound state relic 1.1 TeV"}	Found 8 papers matching query: 'wino higgsino thermal mass Sommerfeld bound state relic 1.1 TeV' (source: OpenAlex) Paper 1: arXiv ID: 1010.2172 Title: Relic densities including Sommerfeld enhancemen...
15	hep-research-aid search_arxiv	{"max_results": 8, "query": "LZ high-energy nuclear recoil search inelastic dark matter 2026"}	Found 8 papers matching query: 'LZ high-energy nuclear recoil search inelastic dark matter 2026' (source: OpenAlex) Paper 1: arXiv ID: 2609.10491 Title: Inelastic Dark Matter and High-Energy Recoil S...
16	hep-research-aid search_arxiv	{"max_results": 6, "query": "inelastic higgsino solar capture IceCube neutrino bound"}	Found 6 papers matching query: 'inelastic higgsino solar capture IceCube neutrino bound' (source: OpenAlex) Paper 1: arXiv ID: 2609.02775 Title: Strong Constraints on Higgsino Dark Matter from Solar ...
17	hep-research-aid search_arxiv	{"max_results": 5, "query": "higgsino nucleon elastic cross section loop neutrino fog"}	Found 5 papers matching query: 'higgsino nucleon elastic cross section loop neutrino fog' (source: OpenAlex) Paper 1: arXiv ID: 2609.01504 Title: Higgsino Above the Sea of Fog Authors: JiJi Fan, Matt...

#	Tool	Arguments	Outcome
18	hep-research-aid search_arxiv	{"max_results": 5, "query": "LZ collaboration search high energy nuclear recoils 2.84 tonne-year"}	Found 5 papers matching query: 'LZ collaboration search high energy nuclear recoils 2.84 tonne-year' (source: OpenAlex) Paper 1: arXiv ID: 2609.02608 Title: Dark Matter at the Kinematic Edge: Interpr...
19	engine describe_tools	{}	{ "server": "inelastic-dm-server", "description": "Bridges particle physics (WIMP mass, mass splitting, cross section) with cosmological observables (relic density, recoil spectra, indirect detect...
20	engine describe_tools	{}	{ "server": "inelastic-dm-server", "description": "Bridges particle physics (WIMP mass, mass splitting, cross section) with cosmological observables (relic density, recoil spectra, indirect detect...
21	serve-idm.sh predict_recoil	{"m_chi_GeV": 1100, "delta_keV": 300}	{ "parameters": { "m_chi_GeV": 1100.0, "delta_keV": 300.0, "sigma_n_cm2": 7.429930732553727e-39, "is_higgsino": true }, "kinematics": { "E_R_peak_keV": 270.0, "v_min_at_p...
22	serve-idm.sh compute_relic_density	{"m_chi_GeV": 900}	{ "parameters": { "m_chi_GeV": 900.0, "is_higgsino": true, "sigma_v_cm3_s": 3.7345679012345686e-26 }, "relic_density": { "omega_h2": 0.0731, "planck_value": 0.12, "planck...

#	Tool	Arguments	Outcome
23	serve-idm.sh predict_recoil	{"m_chi_GeV": 1100, "delta_keV": 380}	{ "parameters": { "m_chi_GeV": 1100.0, "delta_keV": 380.0, "sigma_n_cm2": 7.429930732553727e-39, "is_higgsino": true }, "kinematics": { "E_R_peak_keV": 342.1, "v_min_at_p...
24	serve-idm.sh compute_relic_density	{"m_chi_GeV": 1000}	{ "parameters": { "m_chi_GeV": 1000.0, "is_higgsino": true, "sigma_v_cm3_s": 3.0250000000000006e-26 }, "relic_density": { "omega_h2": 0.0898, "planck_value": 0.12, "planc...
25	serve-idm.sh predict_recoil	{"m_chi_GeV": 1100, "delta_keV": 500}	{ "parameters": { "m_chi_GeV": 1100.0, "delta_keV": 500.0, "sigma_n_cm2": 7.429930732553727e-39, "is_higgsino": true }, "kinematics": { "E_R_peak_keV": 450.1, "v_min_at_p...
26	serve-idm.sh compute_relic_density	{"m_chi_GeV": 1100}	{ "parameters": { "m_chi_GeV": 1100.0, "is_higgsino": true, "sigma_v_cm3_s": 2.5e-26 } , "relic_density": { "omega_h2": 0.1083, "planck_value": 0.12, "planck_error": 0.001...
27	serve-idm.sh predict_recoil	{"m_chi_GeV": 1100, "delta_keV": 566}	{ "parameters": { "m_chi_GeV": 1100.0, "delta_keV": 566.0, "sigma_n_cm2": 7.429930732553727e-39, "is_higgsino": true }, "kinematics": { "E_R_peak_keV": 509.5, "v_min_at_p...

#	Tool	Arguments	Outcome
28	serve-idm.sh compute_relic_density	{"m_chi_GeV": 1200}	{ "parameters": { "m_chi_GeV": 1200.0, "is_higgsino": true, "sigma_v_cm3_s": 2.1006944444444443e-26 }, "relic_density": { "omega_h2": 0.1285, "planck_value": 0.12, "planc...
29	serve-idm.sh predict_recoil	{"m_chi_GeV": 1100, "delta_keV": 700}	{ "parameters": { "m_chi_GeV": 1100.0, "delta_keV": 700.0, "sigma_n_cm2": 7.429930732553727e-39, "is_higgsino": true }, "kinematics": { "E_R_peak_keV": 630.1, "v_min_at_p...
30	serve-idm.sh compute_relic_density	{"m_chi_GeV": 1300}	{ "parameters": { "m_chi_GeV": 1300.0, "is_higgsino": true, "sigma_v_cm3_s": 1.789940828402367e-26 }, "relic_density": { "omega_h2": 0.1503, "planck_value": 0.12, "planck...
31	serve-idm.sh check_all_constraints	{"m_chi_GeV": 1100, "delta_keV": 400}	{ "parameters": { "m_chi_GeV": 1100.0, "delta_keV": 400.0, "sigma_n_cm2": 7.429930732553727e-39, "is_higgsino": true } , "relic_density": { "omega_h2": 0.1083, "tension_si...
32	serve-idm.sh check_all_constraints	{"m_chi_GeV": 1159, "delta_keV": 400}	{ "parameters": { "m_chi_GeV": 1159.0, "delta_keV": 400.0, "sigma_n_cm2": 7.430576349889478e-39, "is_higgsino": true }, "relic_density": { "omega_h2": 0.12, "tension_sigm...

#	Tool	Arguments	Outcome
33	serve-idm.sh check_all_ constraints	{"m_chi_GeV": 1100, "delta_ keV": 566}	{ "parameters": { "m_chi_GeV": 1100.0, "delta_keV": 566.0, "sigma_n_cm2": 7.429930732553727e-39, "is_higgsino": true } , "relic_density": { "omega_h2": 0.1083, "tension_si...
34	serve-idm.sh check_all_ constraints	{"m_chi_GeV": 1100, "delta_ keV": 1000}	{ "parameters": { "m_chi_GeV": 1100.0, "delta_keV": 1000.0, "sigma_n_cm2": 7.429930732553727e-39, "is_higgsino": true } , "relic_density": { "omega_h2": 0.1083, "tension_s...
35	serve-idm.sh predict_recoil	{"delta_keV": 380, "m_chi_ GeV": 1000}	{ "parameters": { "m_chi_GeV": 1000.0, "delta_keV": 380.0, "sigma_n_cm2": 7.428662722453626e-39, "is_higgsino": true }, "kinematics": { "E_R_ peak_keV": 338.7, "v_min_ at_p...
36	serve-idm.sh predict_recoil	{"delta_keV": 380, "m_chi_ GeV": 1200}	{ "parameters": { "m_chi_GeV": 1200.0, "delta_keV": 380.0, "sigma_n_cm2": 7.430987655656208e-39, "is_higgsino": true }, "kinematics": { "E_R_ peak_keV": 344.9, "v_min_ at_p...
37	serve-idm.sh predict_recoil	{"delta_keV": 350, "m_chi_ GeV": 1100}	{ "parameters": { "m_chi_GeV": 1100.0, "delta_keV": 350.0, "sigma_n_cm2": 7.429930732553727e-39, "is_higgsino": true }, "kinematics": { "E_R_ peak_keV": 315.1, "v_min_ at_p...

#	Tool	Arguments	Outcome
38	serve-idm.sh scan_mass_splitting	{"m_chi_min": 800, "m_chi_max": 1400, "delta_min": 0, "delta_max": 800, "n_points": 17}	{ "scan_file": "/Users/nramachandra/github-integration/inelastic-dm-server/output/scan_idm.npz", "grid": "17 x 17 = 289 points", "ranges": { "m_chi_GeV": [800.0, 1400.0], ...
39	serve-idm.sh plot_recoil_spectrum	{"m_chi_GeV": 1100, "delta_keV": 380}	{ "plot_file": "/Users/nramachandra/github-integration/inelastic-dm-server/output/recoil_spectrum.png", "parameters": { "m_chi_GeV": 1100.0, "delta_keV": 380.0 } }
40	serve-idm.sh predict_annual_modulation	{"m_chi_GeV": 1100, "delta_keV": 350, "E_R_keV": 248}	{ "parameters": { "m_chi_GeV": 1100.0, "delta_keV": 350.0, "E_R_keV": 248.0 }, "modulation": { "peak_day": 153, "peak_date": "June 02", "trough_day": 336, "trough_dat...
41	board post_finding	{"claim": "The inelastic-DM server's compute_relic_density (is_higgsino=true) gives Omega h^2 = 0.0731, 0.0898, 0.1083, 0.1285, 0.1503 at m_chi = 900, 1000, 1100, 1200, 1300 GeV. Linear interpolation crosses Planck 0.120...	Recorded finding f-001.
42	board post_finding	{"claim": "check_all_constraints (is_higgsino) at m=1100 GeV with delta = 400, 566 and 1000 keV, and at m=1159 GeV with delta=400 keV, returns VIABLE every time. Fermi-LAT and CMB are not excluded, sigma_n = 7.43e-39 cm^...	Recorded finding f-002.

#	Tool	Arguments	Outcome
43	board post_question	{"text": "Why does check_all_constraints return N_expected_LZ = 0.00 for a 1.1 TeV higgsino at delta = 400 keV (sigma_n = 7.4e-39 cm^2) when the literature finds about 1 event at 340-380 keV? Is the velocity cutoff or en...	Recorded question q-06.
44	board post_finding	{"claim": "With the inelastic-DM server (Z-exchange higgsino, sigma_n = 7.43e-39 cm^2, LZ 2.84 t-yr extended NR window, v_esc+v_E = 776 km/s), the expected LZ count at m_chi = 1100 GeV is 744.6 at delta = 300 keV, 1.97 a...	Recorded finding f-003.
45	board post_finding	{"claim": "Xenon's kinematic reach at the thermal higgsino mass is delta_max = mu v_max^2/2. With m_Xe of about 122 GeV, m_chi = 1100 GeV gives mu = 109.8 GeV. For v_max = 776 km/s (the tool's value, beta^2 = 6.70e-6) th...	Recorded finding f-004.
46	board post_finding	{"claim": "The scan over m_chi = 800-1400 GeV and delta = 0-800 keV (17x17 grid) finds 144 of 289 points passing all constraints and 9 points that predict 0.3-5 LZ events. The 1100 GeV, 380 keV recoil-spectrum plot is at...	Recorded finding f-005.
47	serve-idm.sh plot_constraint_summary	{"scan_file": "/Users/nramachandra/github-integration/inelastic-dm-server/output/scan_idm.npz"}	{ "plot_file": "/Users/nramachandra/github-integration/inelastic-dm-server/output/constraint_summary.png", "best_fit": { "m_chi_GeV": 988.0, "delta_keV": 350.0 } }

#	Tool	Arguments	Outcome
48	board post_finding	{"claim": "plot_constraint_ summary on the 17x17 scan (m_chi = 800-1400 GeV, delta = 0-800 keV) wrote /Users/nramachandra/ github-integration/ inelastic-dm-server/output/ constraint_summary.png and reports the best-fit point...	Recorded finding f-006.
49	local get_arxiv_ metadata	{"arxiv_id": "1807.06209"}	Paper 1807.06209 not found
50	local get_arxiv_ metadata	{"arxiv_id": "2410.15733"}	Paper 2410.15733 not found
51	local download_arxiv_ full_text	{"arxiv_id": "1807.06209", "output_dir": "/ Users/nramachandra/ hep-genesis-testing/ projects/ ramp-combo-higgsino-dd/runs/ 261016_143117"}	Download failed: Paper 1807.06209 not found
52	local download_arxiv_ full_text	{"arxiv_id": "2410.15733", "output_dir": "/ Users/nramachandra/ hep-genesis-testing/ projects/ ramp-combo-higgsino-dd/runs/ 261016_143117"}	Download failed: Paper 2410.15733 not found
53	hep-research-aid download_full_ arxiv_paper	{"paper_id": "1807.06209", "output_dir": "/ Users/nramachandra/ hep-genesis-testing/ projects/ ramp-combo-higgsino-dd/runs/ 261016_143117"}	Error: Error downloading/extracting paper: 'Search' object has no attribute 'results'
54	hep-research-aid search_arxiv	{"query": "higgsino dark matter thermal relic Sommerfeld 1.1 TeV", "max_ results": 5}	Found 5 papers matching query: 'higgsino dark matter thermal relic Sommerfeld 1.1 TeV' (source: OpenAlex) Paper 1: arXiv ID: 1404.0682 Title: Neutralino dark matter at 14 TeV and 100 TeV Authors: Mat...

#	Tool	Arguments	Outcome
55	local cite	{"arxiv_id": "2410.15733", "note": "[\"in which the observed DM abundance is realized with the Higgsino mass of ~1.2 TeV\"]", "bib_path": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-higgsino-dd/runs/2610...	METADATA FAILED: arxiv_id 2410.15733 not found
56	hep-research-aid search_arxiv	{"max_results": 5, "query": "Planck 2018 results VI cosmological parameters"}	Found 5 papers matching query: 'Planck 2018 results VI cosmological parameters' (source: OpenAlex) Paper 1: arXiv ID: 1907.12875 Title: Planck 2018 results Authors: N. Aghanim, Y. Akrami, M. Ashdown,...
57	hep-research-aid search_arxiv	{"max_results": 5, "query": "XENONnT first WIMP search nuclear recoil spin-independent limit"}	Found 5 papers matching query: 'XENONnT first WIMP search nuclear recoil spin-independent limit' (source: OpenAlex) Paper 1: arXiv ID: 2303.14729 Title: First Dark Matter Search with Nuclear Recoils ...
58	board post_finding	{"claim": "The pre-2026 Sommerfeld- and coannihilation-corrected thermal pure-higgsino mass is about 1.2 TeV. The abstract of arXiv:2410.15733 (Fukuda, Li, Moroi, Niki, JHEP06(2025)091, DOI 10.1007/jhep06(2025)091) says:...	Recorded finding f-007.
59	board post_finding	{"claim": "XENONnT's 3.1 tonne-year elastic WIMP search (arXiv:2502.18005, DOI 10.1103/msw4-t342) found no significant excess. Its spin-independent limit has a minimum of 1.7e-47 cm^2 (90% CL) at 30 GeV, with best median...	Recorded finding f-008.

#	Tool	Arguments	Outcome
60	board post_dead_end	{"what": "Verbatim citation of 1807.06209 (Planck 2018 VI) and 2410.15733 using get_arxiv_metadata, download_arxiv_full_text, download_full_arxiv_ paper and cite.", "why": "During this run, every arXiv-API-backed tool ret...	Recorded dead end x-01.
61	hep-research-aid search_arxiv	{"query": "Planck 2018 results cosmological parameters dark matter density 0.120 Hubble constant 67.4", "max_ results": 3}	Found 3 papers matching query: 'Planck 2018 results cosmological parameters dark matter density 0.120 Hubble constant 67.4' (source: OpenAlex) Paper 1: arXiv ID: 2403.14247 Title: Hint of Dark Matter...
62	board post_question	{"text": "Can a later run, once the arXiv API is back, retrieve and cite the verbatim Planck 2018 VI value Omega_c h^2 = 0.120 +/ - 0.001 (arXiv:1807.06209) ? Can it also obtain the XENONnT/LZ spin-independent limit at 1 T...	Recorded question q-07.
63	serve-idm.sh predict_recoil	{"m_chi_GeV": 1100, "delta_ keV": 355}	{ "parameters": { "m_chi_GeV": 1100.0, "delta_keV": 355.0, "sigma_n_cm2": 7.429930732553727e-39, "is_higgsino": true }, "kinematics": { "E_R_ peak_keV": 319.6, "v_min_ at_p...
64	serve-idm.sh predict_recoil	{"m_chi_GeV": 1100, "delta_ keV": 360}	{ "parameters": { "m_chi_GeV": 1100.0, "delta_keV": 360.0, "sigma_n_cm2": 7.429930732553727e-39, "is_higgsino": true }, "kinematics": { "E_R_ peak_keV": 324.1, "v_min_ at_p...

#	Tool	Arguments	Outcome
65	serve-idm.sh predict_recoil	{"m_chi_GeV": 1100, "delta_keV": 365}	{ "parameters": { "m_chi_GeV": 1100.0, "delta_keV": 365.0, "sigma_n_cm2": 7.429930732553727e-39, "is_higgsino": true }, "kinematics": { "E_R_peak_keV": 328.6, "v_min_at_p...
66	serve-idm.sh predict_recoil	{"m_chi_GeV": 1100, "delta_keV": 370}	{ "parameters": { "m_chi_GeV": 1100.0, "delta_keV": 370.0, "sigma_n_cm2": 7.429930732553727e-39, "is_higgsino": true }, "kinematics": { "E_R_peak_keV": 333.1, "v_min_at_p...
67	serve-idm.sh predict_recoil	{"m_chi_GeV": 1100, "delta_keV": 375}	{ "parameters": { "m_chi_GeV": 1100.0, "delta_keV": 375.0, "sigma_n_cm2": 7.429930732553727e-39, "is_higgsino": true }, "kinematics": { "E_R_peak_keV": 337.6, "v_min_at_p...
68	serve-idm.sh predict_recoil	{"m_chi_GeV": 1159, "delta_keV": 360}	{ "parameters": { "m_chi_GeV": 1159.0, "delta_keV": 360.0, "sigma_n_cm2": 7.430576349889478e-39, "is_higgsino": true }, "kinematics": { "E_R_peak_keV": 325.7, "v_min_at_p...
69	hep-research-aid search_arxiv	{"query": "H.E.S.S. higgsino gamma-ray line limit Galactic Center", "max_results": 6}	Found 6 papers matching query: 'H.E.S.S. higgsino gamma-ray line limit Galactic Center' (source: OpenAlex) Paper 1: arXiv ID: 2608.07234 Title: Search for Gamma-Ray Spectral Lines from Dark Matter An...

#	Tool	Arguments	Outcome
70	hep-research-aid search_arxiv	{"query": "thermal higgsino gamma-ray line Sommerfeld 1.1 TeV indirect detection", "max_results": 6}	Found 6 papers matching query: 'thermal higgsino gamma-ray line Sommerfeld 1.1 TeV indirect detection' (source: OpenAlex) Paper 1: arXiv ID: 2609.01590 Title: TeV Higgsino Dark Matter from LZ Nuclear...
71	local get_arxiv_ metadata	{"arxiv_id": "2609.02823"}	Paper 2609.02823 not found
72	hep-research-aid search_arxiv	{"query": "CTA higgsino sensitivity Cherenkov Telescope Array wino higgsino", "max_results": 6}	Found 6 papers matching query: 'CTA higgsino sensitivity Cherenkov Telescope Array wino higgsino' (source: OpenAlex) Paper 1: arXiv ID: 2008.00692 Title: Prospects for detecting heavy WIMP dark matte...
73	board post_finding	{"claim": "With predict_ recoil (Z-exchange higgsino, sigma_n = 7.43e-39 cm^ 2, LZ 2.84 t-yr extended NR window, hard cutoff v_ esc+v_E = 776 km/s), m_chi = 1100 GeV gives N_LZ = 1.97, 0.494, 0.085, 0.009, 0.00, 0.00 at del...	Recorded finding f-009.
74	board post_finding	{"claim": "This answers q-06. predict_recoil has no halo parameters (inputs: m_ chi_GeV, delta_keV, sigma_ n_cm2, is_higgsino), so v_ esc cannot be varied. The N_LZ = 0 at delta >= 370 keV comes from the tool's hard 776 km/...	Recorded finding f-010.
75	local get_arxiv_ metadata	{"arxiv_id": "2609.02775"}	Paper 2609.02775 not found
76	local download_arxiv_ full_text	{"arxiv_id": "2405.13104", "output_dir": "/ Users/nramachandra/ hep-genesis-testing/ projects/ ramp-combo-higgsino-dd/runs/ 261016_143117"}	Download failed: Paper 2405.13104 not found

#	Tool	Arguments	Outcome
77	board post_finding	{"claim": "The indirect-detection status of the about 1.1 TeV thermal higgsino, taken from abstracts only: it is not excluded by current gamma-ray searches, and the result depends on the halo profile. (a) arXiv: 2405.1310...	Recorded finding f-011.
78	board post_finding	{"claim": "CTAO is the decisive future test. arXiv:2008.00692 (Rinchiuso et al., PRD 103, 023011) finds that CTA 'has the potential to become the first experiment that has sensitivity to the thermal' higgsino. arXiv:2405...	Recorded finding f-012.
79	local get_arxiv_ metadata	{"arxiv_id": "1807.06209"}	Paper 1807.06209 not found
80	local get_arxiv_ metadata	{"arxiv_id": "2410.15733"}	Paper 2410.15733 not found
81	hep-research-aid download_full_ arxiv_paper	{"paper_id": "2609.02823", "output_dir": "/ Users/nramachandra/ hep-genesis-testing/ projects/ ramp-combo-higgsino-dd/runs/ 261016_143117"}	Error: Error downloading/extracting paper: 'Search' object has no attribute 'results'
82	local download_arxiv_ full_text	{"arxiv_id": "1807.06209", "output_dir": "/ Users/nramachandra/ hep-genesis-testing/ projects/ ramp-combo-higgsino-dd/runs/ 261016_143117"}	Download failed: Paper 1807.06209 not found
83	hep-research-aid search_arxiv	{"query": "Planck 2018 results VI cosmological parameters", "max_results": 5}	Found 5 papers matching query: 'Planck 2018 results VI cosmological parameters' (source: OpenAlex) Paper 1: arXiv ID: 1907.12875 Title: Planck 2018 results Authors: N. Aghanim, Y. Akrami, M. Ashdown,...

#	Tool	Arguments	Outcome
84	local cite	{"arxiv_id": "2410.15733", "note": "[\"in which the observed DM abundance is realized with the Higgsino mass of ~1.2 TeV\"]", "bib_path": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-higgsino-dd/runs/2610...	METADATA FAILED: arxiv_id 2410.15733 not found
85	hep-research-aid search_arxiv	{"max_results": 5, "query": "LUX-ZEPLIN extended nuclear recoil energy window 248 keV"}	Found 5 papers matching query: 'LUX-ZEPLIN extended nuclear recoil energy window 248 keV' (source: OpenAlex) Paper 1: arXiv ID: 2609.02823 Title: Search for dark matter particle interactions in an ex...
86	hep-research-aid search_arxiv	{"max_results": 5, "query": "higgsino spin-independent cross section loop level neutrino floor xenon"}	Found 5 papers matching query: 'higgsino spin-independent cross section loop level neutrino floor xenon' (source: OpenAlex) Paper 1: arXiv ID: 2203.02309 Title: A next-generation liquid xenon observa...
87	board post_finding	{"claim": "The OpenAlex record for arXiv:2609.02823 (LZ, 'Search for dark matter particle interactions in an extended nuclear recoil energy window with the LUX-ZEPLIN (LZ) experiment', 2026-09-02, DOI 10.48550/arxiv.2609...	Recorded finding f-013.
88	board post_finding	{"claim": "Abstract-level literature anchors from OpenAlex. (a) arXiv: 2403.19598 (S. P. Martin, PRD 109, 095045, 2024) says a thermal higgsino-like DM \"mass must be between about 100 GeV and 1150 GeV\". (b) arXiv:2607.2...	Recorded finding f-014.

#	Tool	Arguments	Outcome
89	board post_dead_end	{"what": "Verification and cite() of 2609.02823, 2609.02775, 1807.06209 and 2410.15733 through get_arxiv_metadata, download_arxiv_full_text, download_full_arxiv_paper and cite (wave 2, second attempt after x-01).", "why"...	Recorded dead end x-02.

B Research board

The shared board of the research runs in this session, generated mechanically: what the team set out to investigate, what it found (with the tool calls that are the evidence), what failed, and how the critic judged each wave.

B.1 Directions

Id	Status	Goal	Outcome
d-01	closed	Compute the thermal relic density Omega h^2 of a higgsino-like (inelastic) DM candidate at m = 0.9, 1.0, 1.1, 1.2 and 1.3 TeV with the inelastic-DM tools. Compare each value with Planck (Omega h^2 = 0.120 +/- 0.001) and locate the mass where the tool's relic curve crosses the Planck abundance.	Relic densities for a higgsino-like dark matter candidate (compute_relic_density with is_higgsino=true; the tool sets the annihilation cross section automatically): Omega h^2 = 0.0731, 0.0898, 0.1083, 0.1285 and 0.1503 at m = 900, 1000, 1100, 1200 and 1300 GeV (tc-...-23/25/27/29/31). The curve reaches the Planck value of 0.120 at about 1158 GeV, +/-6 GeV from the server's +/-0.0012 Planck band (t
d-02	closed	Place the thermal higgsino (m about 1.0-1.2 TeV) relative to LZ/XENONnT. Compute inelastic recoil spectra and expected event counts for delta = 300-600 keV, scan the (m_chi, delta) plane against all constraints, and identify where the kinematic reach of xenon ends.	The inelastic-DM server models the higgsino with Z exchange (sigma_n = 7.43e-39 cm^2), the LZ 2.84 t-yr extended nuclear-recoil window, and a halo cut-off of v_esc + v_E = 776 km/s (tc-...-20, -21). At m_chi = 1100 GeV, the expected LZ event counts are: 744.6 at delta = 300 keV (grossly excluded against 1 observed event); 1.97 at 350 keV (P(N>=1) = 0.86, peak recoil energy E_R = 315 keV); and 0.00

Id	Status	Goal	Outcome
d-03	blocked	Verify and cite the key literature: the higgsino relic crossing mass, the LZ 2.84 t-yr extended-window result, the solar-capture delta bound, the elastic loop-level cross section relative to the XENONnT limits and the neutrino fog, and the Planck Omega h^2. Obtain verbatim quotes.	No cite() entries could be made, so the success criterion of at least 5 cited quotes was not met. The arXiv-API layer is still down. get_arxiv_metadata returned 'not found' for 2609.02823, 2609.02775, 1807.06209 and 2410.15733, including the well-established Planck ID, so these results say nothing about whether the papers exist. download_arxiv_full_text failed for 1807.06209, download_full_arxiv_p
d-04	blocked	Cite the established, non-2026 baseline: the Planck 2018 Omega_c h^2, the Sommerfeld-corrected pure-higgsino thermal mass from pre-2026 literature, and the XENONnT/LZ elastic SI limit at about 1 TeV compared with the loop-level higgsino cross section and the neutrino fog.	Two of the three anchors were confirmed from abstracts returned by the OpenAlex-backed search_arxiv tool. None could be added to references.bib with cite(), so the success criterion of 3 cite() entries was not met. (1) Thermal higgsino mass, finding f-007: arXiv:2410.15733 (Fukuda, Li, Moroi, Niki, JHEP06(2025)091, DOI 10.1007/jhep06(2025)091) states 'in which the observed DM abundance is realized
d-05	closed	Map, with the inelastic-DM tools, the mass splitting delta at which a thermal higgsino gives about 1 and about 0.3 expected LZ events. Test how sensitive that window is to the halo velocity cutoff, then compare it with the literature fits: 339 keV (2609.26698), about 350 keV (2609.01504) and 377 keV (2609.02608).	I mapped the window with predict_recoil (Z-exchange higgsino, sigma_n = 7.43e-39 cm^2, LZ 2.84 t-yr extended NR window, hard cutoff v_esc+v_E = 776 km/s). At m_chi = 1100 GeV the expected LZ count is N_LZ = 1.97, 0.494, 0.085, 0.009, 0.00 and 0.00 at delta = 350, 355, 360, 365, 370 and 375 keV, with the spectrum peaking at E_R = 315-338 keV (tc-...-65 to -69). Interpolating in log N gives delta(N=

Id	Status	Goal	Outcome
d-06	closed	Establish the current indirect-detection status of the 1.0-1.2 TeV thermal higgsino from the literature: the H.E.S.S., Fermi-LAT and MAGIC gamma-ray line and continuum limits against the Sommerfeld-enhanced s-wave annihilation cross section, and the CTAO projections. This replaces the server's p-wave treatment, which is physically wrong for a higgsino.	Verdict, based on abstracts only (download_arxiv_full_text on 2405.13104 returned 'not found', tc-261016_1-w2-d-06-76): the 1.0-1.2 TeV thermal higgsino is not excluded by current gamma-ray line or continuum searches, and the result depends on the halo profile. Current searches: arXiv:2405.13104 (Rodd, Safdi, Xu, PRD 110, 043003) ran a 15-yr Fermi-LAT line-like search that comes 'within an order o

B.2 Findings and evidence

Id	Confidence	Claim	Evidence (tool calls, figures)
f-001	medium	The inelastic-DM server's compute_relic_density (is_higgsino=true) gives Omega h^2 = 0.0731, 0.0898, 0.1083, 0.1285, 0.1503 at m_chi = 900, 1000, 1100, 1200, 1300 GeV. Linear interpolation crosses Planck 0.120 at m about 1158 GeV (+/-6 GeV for the +/-0.0012 band). A direct check at 1159 GeV returns Omega h^2 = 0.120. This lies within the 1.08-1.2 TeV literature range but about 60-80 GeV above the 1.08-1.1 TeV values.	tc-261016_1-w1-d-01-23, tc-261016_1-w1-d-01-25, tc-261016_1-w1-d-01-27, tc-261016_1-w1-d-01-29, tc-261016_1-w1-d-01-31, ...
f-002	medium, **contested**	check_all_constraints (is_higgsino) at m=1100 GeV with delta = 400, 566 and 1000 keV, and at m=1159 GeV with delta=400 keV, returns VIABLE every time. Fermi-LAT and CMB are not excluded, sigma_n = 7.43e-39 cm^2, and N_expected_LZ = 0.00 at every point. N_LZ = 0 at delta = 400 keV contradicts the literature, which reports about 1 event at delta about 340-380 keV. The server's present-day sigma_v (about 7e-42 to 5e-41 cm^3/s, labelled 'p-wave + mass-splitting suppressed') is also far below the s-wave higgsino value of about 1e-26 cm^3/s.	tc-261016_1-w1-d-01-32, tc-261016_1-w1-d-01-33, tc-261016_1-w1-d-01-34, tc-261016_1-w1-d-01-35

Id	Confidence	Claim	Evidence (tool calls, figures)
f-003	high	With the inelastic-DM server (Z-exchange higgsino, sigma_n = 7.43e-39 cm^2, LZ 2.84 t-yr extended NR window, v_esc+v_E = 776 km/s), the expected LZ count at m_chi = 1100 GeV is 744.6 at delta = 300 keV, 1.97 at 350 keV (P(N>=1) = 0.86, peak E_R = 315 keV), and 0.00 at 380, 500, 566 and 700 keV (peak E_R = 342, 450, 510, 630 keV). The 248 keV event needs v_min = 762 km/s at 350 keV (accessible) and 799 km/s at 380 keV (not accessible).	tc-261016_1-w1-d-02-20, tc-261016_1-w1-d-02-21, tc-261016_1-w1-d-02-38, tc-261016_1-w1-d-02-24, tc-261016_1-w1-d-02-26, ...
f-004	high	Xenon's kinematic reach at the thermal higgsino mass is delta_max = mu v_max^2/2. With m_Xe of about 122 GeV, m_chi = 1100 GeV gives mu = 109.8 GeV. For v_max = 776 km/s (the tool's value, beta^2 = 6.70e-6) this gives delta_max of about 368 keV; for v_max = 794 km/s (544+250) it gives about 385 keV. This agrees with the tool: 350 keV is accessible and 380 keV gives 0 events. At delta = 380 keV the 248 keV v_min is 802, 799 and 796 km/s for m = 1000, 1100 and 1200 GeV, all inaccessible, so the reach depends only weakly on mass.	tc-261016_1-w1-d-02-36, tc-261016_1-w1-d-02-37, tc-261016_1-w1-d-02-24, tc-261016_1-w1-d-02-38
f-005	medium	The scan over m_chi = 800-1400 GeV and delta = 0-800 keV (17x17 grid) finds 144 of 289 points passing all constraints and 9 points that predict 0.3-5 LZ events. The 1100 GeV, 380 keV recoil-spectrum plot is at /Users/nramachandra/github-integration/inelastic-dm-server/output/recoil_spectrum.png. The annual modulation at 1100 GeV, 350 keV and E_R = 248 keV has a fractional amplitude of 0.985, peaking on June 2; the event date of September 1 sees 22.5% of the peak rate.	tc-261016_1-w1-d-02-39, tc-261016_1-w1-d-02-40, tc-261016_1-w1-d-02-41
f-006	high	plot_constraint_summary on the 17x17 scan (m_chi = 800-1400 GeV, delta = 0-800 keV) wrote /Users/nramachandra/github-integration/inelastic-dm-server/output/constraint_summary.png and reports the best-fit point at m_chi = 988 GeV, delta = 350 keV.	tc-261016_1-w1-d-02-48, tc-261016_1-w1-d-02-39

Id	Confidence	Claim	Evidence (tool calls, figures)
f-007	high, **contested**	The pre-2026 Sommerfeld- and coannihilation-corrected thermal pure-higgsino mass is about 1.2 TeV. The abstract of arXiv:2410.15733 (Fukuda, Li, Moroi, Niki, JHEP06(2025)091, DOI 10.1007/jhep06(2025)091) says: "in which the observed DM abundance is realized with the Higgsino mass of ~1.2 TeV".	tc-261016_1-w1-d-04-54
f-008	medium, **contested**	XENONnT's 3.1 tonne-year elastic WIMP search (arXiv:2502.18005, DOI 10.1103/msw4-t342) found no significant excess. Its spin-independent limit has a minimum of 1.7e-47 cm^2 (90% CL) at 30 GeV, with best median sensitivity 1.4e-47 cm^2 at 41 GeV. The earlier 1.09 t-yr result (arXiv:2303.14729) reached 2.58e-47 cm^2 at 28 GeV. The abstracts do not give the limit at 1 TeV. Because the limit scales roughly linearly with mass at high mass, it should be about 1e-45 cm^2 near 1 TeV, but that is an inference, not a quoted value.	tc-261016_1-w1-d-04-57
f-009	high	With predict_recoil (Z-exchange higgsino, sigma_n = 7.43e-39 cm^2, LZ 2.84 t-yr extended NR window, hard cutoff v_esc+v_E = 776 km/s), m_chi = 1100 GeV gives N_LZ = 1.97, 0.494, 0.085, 0.009, 0.00, 0.00 at delta = 350, 355, 360, 365, 370, 375 keV (peak E_R = 315, 320, 324, 329, 333, 338 keV). Log-linear interpolation gives delta(N=1) = 352.5 keV and delta(N=0.3) = 356.4 keV, so the window is only about 4 keV wide (N falls about 4-6x per 5 keV). The 248 keV event stops being accessible between 360 and 365 keV (v_min = 774 and 781 km/s). At m_chi = 1159 GeV (the tool's thermal mass), delta = 360	tc-261016_1-w2-d-05-65, tc-261016_1-w2-d-05-66, tc-261016_1-w2-d-05-67, tc-261016_1-w2-d-05-68, tc-261016_1-w2-d-05-69, ...

Id	Confidence	Claim	Evidence (tool calls, figures)
f-010	medium	This answers q-06. predict_recoil has no halo parameters (inputs: m_chi_GeV, delta_keV, sigma_n_cm2, is_higgsino), so v_esc cannot be varied. The N_LZ = 0 at delta >= 370 keV comes from the tool's hard 776 km/s cutoff, which puts delta_max = mu v_max^2/2 at about 368 keV. An analytic rescaling by (794/776)^2 = 1.047 moves the edge to about 385 keV, and moves the tool's delta(N=1) and delta(N=0.3) from 352.5 and 356.4 keV to about 369 and 373 keV. Against the literature: 350 keV (2609.01504) matches the tool directly (N = 1.97). The 776 vs 794 km/s cutoff accounts for about 17 keV of the 24.5 k	tc-261016_1-w2-d-05-65, tc-261016_1-w2-d-05-66, tc-261016_1-w2-d-05-68, tc-261016_1-w2-d-05-69
f-011	medium, **contested**	The indirect-detection status of the about 1.1 TeV thermal higgsino, taken from abstracts only: it is not excluded by current gamma-ray searches, and the result depends on the halo profile. (a) arXiv:2405.13104 (Rodd, Safdi, Xu, PRD 110, 043003, 2024) ran a 15-yr Fermi-LAT line-like search that comes 'within an order one factor' of the sensitivity needed to detect the higgsino, for some Milky Way analog profiles. In that analysis H.E.S.S. is subleading to Fermi for the higgsino. (b) arXiv:2210.03140 (Montanari, Moulin, Rodd, PRD 107, 043028) finds the higgsino 'out of reach by at least a facto	tc-261016_1-w2-d-06-63, tc-261016_1-w2-d-06-71, tc-261016_1-w2-d-06-72
f-012	high, **contested**	CTAO is the decisive future test. arXiv:2008.00692 (Rinchiuso et al., PRD 103, 023011) finds that CTA 'has the potential to become the first experiment that has sensitivity to the thermal' higgsino. arXiv:2405.13104 finds that CTA and SWGO could detect the line-like signal of the 1.1 TeV higgsino 'at high significance... for all but the most pessimistic DM profiles'. arXiv:2506.08084 (PoS ICRC2025) projects that CTAO-North, observing at large zenith angles, could reach the required sensitivity within the next decade. The server's indirect-detection treatment (p-wave, mass-splitting suppressed,	tc-261016_1-w2-d-06-71, tc-261016_1-w2-d-06-72

Id	Confidence	Claim	Evidence (tool calls, figures)
f-013	medium, **contested**	The OpenAlex record for arXiv:2609.02823 (LZ, 'Search for dark matter particle interactions in an extended nuclear recoil energy window with the LUX-ZEPLIN (LZ) experiment', 2026-09-02, DOI 10.48550/arxiv.2609.02823) carries this abstract text: "an exposure of 2.84 tonne-years", "An extended nuclear recoil energy window up to approximately 270 keV", "one event with characteristics consistent with a nuclear recoil of 248±23 (stat)±23 (sys) keV", and "global significance of 2.6σ ... maximum local significance of 3.4σ". The arXiv API itself returns 'not found' for this ID, so the quotes come from	tc-261016_1-w2-d-03-85, tc-261016_1-w2-d-03-64, tc-261016_1-w2-d-03-81
f-014	medium, **contested**	Abstract-level literature anchors from OpenAlex. (a) arXiv:2403.19598 (S. P. Martin, PRD 109, 095045, 2024) says a thermal higgsino-like DM "mass must be between about 100 GeV and 1150 GeV". (b) arXiv:2607.20588 (Bisal et al., 2026) says "NLO corrections lower the DM-nucleon cross section below the neutrino floor" for pure higgsinos, and that a pure higgsino "typically evades current direct detection limits". This abstract quotes no numerical cross section, so the target of about 1e-48 to 1e-49 cm^2 remains unsourced. (c) arXiv:2410.17137 (XLZD Design Book, EPJC 2025) reaches the neutrino fog,	tc-261016_1-w2-d-03-86, tc-261016_1-w2-d-03-83, tc-261016_1-w2-d-03-82, tc-261016_1-w2-d-03-77

B.3 Dead ends

- x-01 (d-04): Verbatim citation of 1807.06209 (Planck 2018 VI) and 2410.15733 using get_arxiv_metadata, download_arxiv_full_text, download_full_arxiv_paper and cite. — During this run, every arXiv-API-backed tool returned 'not found' or failed with an internal error (including HTTP 429). This happened even for well-established IDs, so cite() cannot validate metadata. Only the OpenAlex-backed search_arxiv works, and it returns abstracts only. The Planck VI abstract and the value Omega_c h^2 = 0.120 +/- 0.001 were never retrieved.

- x-02 (d-03): Verification and cite() of 2609.02823, 2609.02775, 1807.06209 and 2410.15733 through get_arxiv_metadata, download_arxiv_full_text, download_full_arxiv_paper and cite (wave 2, second attempt after x-01). — The arXiv API layer still returns 'not found' even for 1807.06209 (Planck 2018 VI), returns HTTP 429 or timeouts, and download_full_arxiv_paper fails with "'Search' object has no attribute 'results"'. cite() fails at the metadata check (2410.15733 'not found'), so no verbatim quote can be validated in this environment. Only OpenAlex abstracts are reachable. A 'not found' result therefore says noth

B.4 Critic verdicts

- `c-01` (wave 0, plan): **revise**; high: Over budget. As written it needs about 15 calls: 1 describe + 5 relic + 3 Sommerfeld toggles + 3 delta checks + 2 check_all. Drop the delta-sensitivity step. Keep the Sommerfeld toggle only if describe_tools shows a switch exists. This brings it to about 9-11 calls.; high: The serve-idm server is probably a generic inelastic-DM model, not a pure higgsino with SU(2) couplings, Sommerfeld enhancement and full coannihilation. If describe_tools does not state higgsino electroweak couplings and Sommerfeld enhancement, label the results as a toy model. In that case, do not ; medium: The delta-sensitivity step (0.3, 0.566, 1.0 MeV) is physically uninformative. A sub-MeV splitting is far below T_f of about m/20, which is about 50 GeV, so the relic density should not depend on delta. If the tool does show delta dependence, that indicates a modelling artifact and should be reported; medium: The hand calculation of delta_max must use the xenon reduced mass, about 110 GeV at m = 1.1 TeV (m_N about 122 GeV), not the DM mass. This gives delta_max of about 0.5 * 110 GeV * (794/3e5)^2, which is about 385 keV. That is already below the LMC-preferred 480-520 keV and below the 566 keV solar bou; medium: The LZ limit built into the tool is probably the earlier WS2022/2024 result, not the 2026 2.84 t-yr extended window. The tool also has no XENONnT limit. State which exposure and energy window the tool uses. Do not attribute tool exclusions to 2609.02823. Binned counts in [14,225), [225,271] and (271; low: Budget is tight, at about 12-13 calls. Drop predict_annual_modulation and the redundant check_all_constraints unless calls remain. Overlay the 566 keV and 340-380 keV literature values only after d-03 confirms that those IDs exist on arXiv.

- `c-02` (wave 1, wave): **continue**; contested `f-002`, `f-007`, `f-008`; medium: The arithmetic checks out: sigma_v = 2.5e-26 x (1100/m)^2 reproduces 3.73e-26 at 900 GeV, and interpolating between 1100 and 1200 GeV gives about 1158 GeV. However, the tool is a single-channel Kolb-Turner scaling calibrated to 1.1 TeV. The 1158 GeV crossing is a calibration artefact, not a higgsino; high: The scan counts 144 of 289 points as viable. That count uses the same relic 'passes' flag that returns True at 7-25 sigma tension, so it probably includes points that are far from the Planck abundance. It also relies on the p-wave-suppressed indirect-detection model, which is wrong for a higgsino. T; medium: The best fit at m = 988 GeV, delta = 350 keV fits the LZ event, not the thermal relic. With the tool's own relic curve, 988 GeV gives Omega h^2 of about 0.087, which is under-abundant. The best fit is therefore not a full-abundance thermal higgsino, and this should be stated.; high: The count falls from 744.6 at 300 keV to 1.97 at 350 keV to 0 at 380 keV. That steep drop comes from the hard cutoff v_esc + v_E = 776 km/s. The literature uses about 544 + 250 km/s, roughly 794 km/s (giving delta_max of about 385 keV), and fits delta of about 377 keV (2609.02608) or 339 keV (2609.2; medium: The elastic leg of the question has not been answered with sourced numbers: the 1 TeV LZ/XENONnT spin-independent limit, the loop-level higgsino cross section, and the neutrino fog. The 2026 IDs (q-03) are still unverified. The indirect-detection conclusion relies on a tool that treats the higgsino

- `c-03` (wave 2, wave): **continue**; contested `f-011`, `f-012`, `f-013`, `f-014`; medium: The (794/776)^2 rescaling of delta(N=1) and delta(N=0.3) from 352.5/356.4 keV to about 369/373 keV is a heuristic. It assumes the whole N(delta) curve stretches with v_max^2, but near the kinematic edge the count depends on how the halo tail is populated between v_min and v_max, not only on where th; low: The 350 keV point (N = 1.97, peak E_R = 315 keV) and the 300 keV point (744.6) come from wave-1 calls under d-02 and are not among this finding's evidence ids. The interpolations check out: ln-linear between 1.97 and 0.494 gives 352.45 keV, and between 0.494 and 0.085 gives 356.4 keV. The finding sh;

high: The board brief still says 'I did not compute any relic densities... No inelastic-DM or relic-density tools were available'. This contradicts d-01, which computed Omega h^2 at five masses (crossing at about 1158 GeV), and d-02 and d-05, which computed LZ event counts. The brief must be rewritten bef; high: The solar-capture bound delta > 566 keV (2609.02775) carries the bottom line: it is the reason the sub-MeV LZ interpretation is disfavoured. Yet no abstract excerpt containing it has been posted. 2609.06760, 2609.15321 and 2610.00226 are also unverified. OpenAlex search_arxiv does work, so abstract-; medium: Neither the elastic loop-level pure-higgsino SI cross section (about 1e-48 to 1e-49 cm^2) nor the xenon elastic limit at 1.1 TeV is sourced. The roughly 1e-45 cm^2 figure is an extrapolation. As a result, the elastic leg of 'where does the higgsino sit relative to current limits' is unsupported.

- `c-04` (wave 2, report): **revise**; high: unsupported: | 900 | 3.73e-26 | 0.0731 | 39.1σ | — The board gives no per-point tension values. The only figures it states are 9.8σ at 1100 GeV and an overall range of 7–25σ. A 39.1σ tension falls outside that range, so the 'as reported by the tool' column is not supported by the ledger. The 25.1σ, ; high: unsupported: | 300 | 744.6 | 270 | 701 | — The board gives N_LZ = 744.6 at 300 keV but no peak E_R (270 keV) and no v_min (701 km/s) for that splitting. Board v_min values exist only for 350 keV (762), 360 keV (774), 365 keV (781) and 380 keV (799).; high: unsupported: | 355 | 0.494 | 320 | 768 | — The v_min of 768 km/s at 355 keV is not on the board. Likewise, v_min = 787 (370 keV), 793 (375 keV), 945 (500 keV), 1025 (566 keV) and 1189 km/s (700 keV) in the same column appear nowhere in the findings or direction results.; high: unsupported: x_f $\approx$ 24.4–24.8 — The board states only x_f of about 24.5. The 24.4–24.8 range is not sourced.; high: unsupported: It also shows how narrow and cutoff-dependent that window is, which explains why the literature fits scatter between 339 and 377 keV. — This is an overclaim. The board says the 339 keV value is a normalization-free sideband optimum and is not comparable to the tool's count. It also says